

A Continuum Capture Certificate from a Legendre Bath Chain for MQM

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Abstract—We formalize the flat-band vacuum reservoir of the MQM port reference using an exact Legendre coordinate transformation. A finite chain enclosure and a propagation-distance bound certify continuum retained error below $3/262144 + 2^{-128} < 2^{-16}$ at $t = 16$. The calculation uses 768 bath coordinates, 240-digit directed intervals, and an exact rational truncation allowance. This closes a conditional continuum point-time target, not finite-temperature operation, full-word execution, physical two-local compilation, or permanent retention. The transformed thermal covariance is specified to prevent independent-reset assumptions from changing the bath model.

Index Terms—Quantum control, continuum reservoirs, orthogonal polynomials, verified propagation, bath-chain mapping.

I. SCOPE AND SOURCE-BOUND INPUTS

The Mayes Quantum Method (MQM) parent port model has four live states $s_0 = pR0$, $s_1 = mR0$, $s_2 = eR0$, and $s_3 = dR1$, plus captured $dR0$ with an emitted photon. The fresh completion flag is already restricted to admitted values. Fixed launch is s_0 with bath vacuum, and all port/bath operations act as identity on an arbitrary bank and spectator. Body background Hamiltonians vanish.

The source spectrum remains $J(\Omega + \delta) = \kappa/(2\pi)$ for $\delta \in [-B, B]$, with $B = 16$, $\Omega = 64$, $\kappa = 8$, and system hopping $(1, 1, \lambda)$, $\lambda^2 = 4\sqrt{2}$. This is the same declared flat vacuum continuum as the preceding finite-reference programme, not the earlier thermal Ohmic spectrum. Exact system–reservoir chain mappings are established prior art [1]; our contribution is the source-specific verification and error ledger.

Finite-grid verification did not close the continuum target: the midpoint operator comparison gave only tB/M . We replace that approximation argument with an exact coordinate map followed by a certified remote-edge truncation. The parameters $B = 16$ and $t = 16$ were already exposed in development; the new chain length and enclosure method were frozen before evaluating this chain. No independent-prediction claim is made.

II. EXACT FLAT-BAND TRIDIAGONALIZATION

Let $b(\delta)$ be continuum bosonic modes with canonical commutator $[b(\delta), b^\dagger(\delta')] = \delta(\delta - \delta')$. Define

$$\phi_n(\delta) = \sqrt{\frac{2n+1}{2B}} P_n(\delta/B), \quad a_n = \int_{-B}^B \phi_n(\delta) b(\delta) d\delta, \quad (1)$$

where P_n is the Legendre polynomial. Orthogonality and completeness in $L^2[-B, B]$ make this an exact unitary bath-coordinate change. In particular $[a_n, a_m^\dagger] = \delta_{nm}$ and vacuum

remains vacuum. This is not a frequency grid or a projection onto 768 physical modes.

The Legendre recurrence gives

$$\delta\phi_n = b_{n+1}\phi_{n+1} + b_n\phi_{n-1}, \quad b_n = \frac{Bn}{\sqrt{4n^2 - 1}}, \quad n \geq 1. \quad (2)$$

The constant continuum coupling has support only on a_0 , with

$$G^2 = \int_{-B}^B J(\Omega + \delta) d\delta = \frac{\kappa B}{\pi} = \frac{128}{\pi}. \quad (3)$$

Hence laboratory bath onsite energies are Ω , while the rotating-frame bath is the Jacobi chain

$$H_B^{\text{rot}} = \sum_{n \geq 1} b_n (a_{n-1}^\dagger a_n + \text{h.c.}), \quad (4)$$

with system entry $G(|s_3\rangle\langle dR0| a_0 + \text{h.c.})$. The vacuum launch and excitation-exchange assumption give an invariant one-excitation sector. The total interacting Hamiltonian conserves energy; off-resonant exchange is not claimed to commute with free subsystem energy.

The full one-excitation problem is therefore one semi-infinite path with weights $(1, 1, \lambda, G, b_1, b_2, \dots)$. The retained system error equals live population

$$q_\infty(t) = \sum_{j=0}^3 |v_j(t)|^2. \quad (5)$$

Tracing the original bath preserves the arbitrary identity spectator. For fixed port launch, q_∞ is also retained half-diamond channel error to the captured target.

III. PROPAGATION-DISTANCE TRUNCATION BOUND

Keep L bath coordinates and cut only link b_L . The omitted tail remains as a disconnected block for comparing unitaries on a common Hilbert space. The finite path has dimension $L + 4$ and its last vertex is distance $D = L + 3$ from launch.

Proposition 1. For $0 \leq t \leq T$, if $\|H_L\| \leq \Lambda$ and $x = \Lambda T/2 < D + 1$, then

$$\|\psi_\infty(t) - \psi_L(t)\| \leq \epsilon_L(T) := \frac{2BT x^D}{D! (1 - x/(D+1))}. \quad (6)$$

Proof. Duhamel at the cut edge gives

$$\|\psi_\infty(t) - \psi_L(t)\| \leq b_L \int_0^t |\langle D | e^{-iH_L s} | 0 \rangle| ds. \quad (7)$$

The Chebyshev expansion of the finite unitary contains coefficients $2(-i)^n J_n(\Lambda s)$ for $n \geq 1$. A degree- n polynomial in a

TABLE I
TRUNCATION DIAGNOSTIC AT $t = 16$

Bath coordinates	One-excitation dimension	Numerical q_L
64	68	0.338405
768	$772 = 4 \times 193$	$1.07125503 \times 10^{-5}$

nearest-neighbor Hamiltonian cannot reach distance D when $n < D$, and $\|T_n(H_L/\Lambda)\| \leq 1$. Poisson's integral [2] gives $|J_n(z)| \leq (|z|/2)^n/n!$ for real z and nonnegative integer n . Consequently

$$|\langle D | e^{-iH_L s} | 0 \rangle| \leq 2 \sum_{n \geq D} \frac{(\Lambda s/2)^n}{n!}. \quad (8)$$

Bounding successive factorial-tail ratios by $x/(D+1)$, taking $s \leq T$, and using $b_L \leq B$ proves the claim. Partial-trace contraction and the pure-state distance inequality give $|q_\infty(t) - q_L(t)| \leq \epsilon_L(T)$. This is a launch-specific propagation bound, not a uniform operator bound for arbitrary bath excitations.

For $L = 768 = 3 \times 2^8$, $T = 16 = 2^4$, and $\Lambda = 24$, the distance is $D = 771 = 3 \times 257$, and $x = 192 = 3 \times 2^6$. Thus

$$\epsilon_{768}(16) \leq \frac{512 \cdot 193 \cdot 192^{771}}{145 \cdot 771!} < 2^{-128}. \quad (9)$$

The last inequality is checked in exact integer/rational arithmetic. The chain basis is Legendre because the spectral density is flat. Chebyshev polynomials are used separately to bound propagation; their weight is not substituted for the bath spectrum.

IV. DIRECTED CONTINUUM CAPTURE CERTIFICATE

The system four-state block has norm below four. Multiplication by δ has norm $B = 16$; the exact bath mapping and every principal truncation preserve this ceiling. Since $G^2 = 128/\pi < 128/3 < 49$, the complete one-excitation path has norm below $16 + 7 = 23 < 24$.

A phase rotation $(-i)^j$ converts the zero-diagonal Hermitian hopping problem into a real skew path. A 240-digit directed Taylor action through degree 1280 encloses the first four live amplitudes at $t = 16$. The unitary integral remainder obeys

$$\|R_{1280}\| \leq \frac{384^{1281}}{1281!}. \quad (10)$$

The resulting squared-norm interval is approximately $1.07125503432923 \times 10^{-5}$ and is strictly below $3/262144$.

Proposition 2 (conditional continuum target). Under the declared flat-band vacuum reference, fixed admitted port launch, and arbitrary identity spectator,

$$q_\infty(16) < \frac{3}{262144} + 2^{-128} < 2^{-16}. \quad (11)$$

Proof. Combine the directed finite-chain enclosure with Proposition 1 and the exact tail inequality. Neither numerical grid agreement nor a Born/Markov approximation enters this certificate.

The 64-coordinate control is cross-checked by dense diagonalization, with vector disagreement below 6.49×10^{-14} . Its

large error cautions against unbounded short-chain approximation. The certified length passes by the analytic truncation allowance and independent intervals. Exact rational polynomial checks comprise 100 inner products and nine link coefficients; these finite checks support implementation, while the all-degree recurrence and completeness establish the mapping.

V. THERMAL INTERFACE AND CLAIM CORRECTIONS

For an equilibrium continuum with inverse temperature β , the transformed covariance is

$$\langle a_n^\dagger a_m \rangle_\beta = \int_{-B}^B \frac{\phi_n(\delta)\phi_m(\delta)}{e^{\beta(\Omega+\delta)} - 1} d\delta. \quad (12)$$

It is generally non-diagonal: preparing independent thermal states on chain sites changes the source bath. For $\beta = 5/16$, $\Omega = 64$, and $B = 16$, spectral occupations obey $n_\beta(\omega) \leq 1/(e^{15} - 1)$. This is a formal occupation bound, not a finite-temperature channel certificate. Thermal excitations remove the vacuum one-excitation restriction; an explicit correlated Gaussian state and multiphoton treatment are required. No material phonon identification is established.

The preceding review requires three scope repairs. The collapsed-resonance floor rejects its declared collective-mode family, not all single-mode resonance. The clock modal audit supports weak terminal overlap for the tested generators; it does not disprove every interference mechanism. A loose continuum bound does not prove grid estimates inaccurate. GR/horizon analogies are not derivations of this MQM bath model.

VI. PROMOTION BOUNDARY AND NEXT GATE

The flat-band vacuum continuum point-time target is now supported by an explicit derivation and replayable certificate. This does not close the earlier thermal Ohmic family, the thirteen-step microscopic word, permanent hold, bath refresh, native joint admission, or physical planar two-local compilation. Bath-chain locality is locality in transformed oscillator coordinates, not a chip-layout theorem. Norm ceilings apply only to the invariant one-excitation sector, not the full bosonic Fock space. Physical promotion remains zero.

Next priorities are a declared finite-temperature continuum construction with preserved covariance and an independent memory-kernel cross-check of the vacuum certificate. The certificate is conditional research evidence; independent theorem review, hardware validation, and comparative advantage remain separate gates.

REFERENCES

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